Evaluating Predictive Utility of Synthetic Liver Cirrhosis Data in IoMT: A Comparative Study of GAN, CTGAN, and TVAE with Consensus-Based Quality Filtering

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***Abstract*—Privacy rules and disease rarity keep clinical datasets small, which limits machine learning for the Internet of Medical Things (IoMT). The Mayo Clinic Primary Biliary Cirrhosis (PBC) data show this: 418 enrolled patients drop to 276 usable records once incomplete cases are removed. We ask whether adding synthetic records to the training set improves prediction on real patients. Three generative models are built from scratch in TensorFlow 2.15 under fixed random seeds: a Vanilla GAN, a conditional tabular GAN (CTGAN), and a Tabular Variational Autoencoder (TVAE). Generated records pass an IQR-based Tukey fence filter, and a consensus step keeps only the records all three models agree on. We measure distributional quality with a tabular Fréchet Inception Distance (FID) and disclosure risk from nearest-neighbour distances. Five scenarios are tested with Random Forest, Gradient Boosting, and Logistic Regression on 83 held-out real patients, using bootstrap intervals, McNemar's exact test, and 5-fold cross-validation that keeps synthetic records on the training side of every fold. CTGAN gives the best FID at 0.0724, and the filtered adversarial pools stay at low disclosure risk (7.3% and 9.6% near-duplicates). Once the evaluation avoids leakage, no scenario beats the real-data baseline: every McNemar test gives $p > 0.05$, all bootstrap intervals overlap, and cross-validation AUC neither rises nor grows more stable. Consensus voting does not lower disclosure risk either, at 21.1%. A classifier trained on CTGAN data alone still reaches AUC 0.82 to 0.84 for Random Forest and Logistic Regression, so synthetic data carries usable signal even where augmentation adds none. Looking similar is not the same as being useful.**

***Index Terms*—Synthetic data generation, generative adversarial networks, variational autoencoder, liver cirrhosis, Internet of Medical Things, predictive utility, consensus voting, IQR filtering, patient privacy.**

I. Introduction

Hospitals and clinics that adopt Internet of Medical Things devices collect continuous streams of clinical measurements from wearable sensors and implanted monitors. Liver enzymes, bilirubin, albumin, and other biomarkers are logged automatically and fed into decision support systems that alert clinicians to patient deterioration [1], [2]. Machine learning models built on such records already predict cirrhosis mortality with useful accuracy [3]. These systems need large, well-labelled training sets to work reliably, and that is where a real problem begins. Privacy laws such as GDPR and HIPAA restrict what patient data can be shared across institutions. On top of this, rare diseases simply do not produce large cohorts. The Mayo Clinic Primary Biliary Cirrhosis trial ran for more than a decade and produced a dataset of 418 records across a randomised arm and a subsequent observational cohort; after removing records with missing measurements, only 276 remain [27], [28]. Building a reliable classifier on 276 patients is genuinely hard.

Synthetic data generation offers a practical way around this. Rather than share real patient records, a generative model learns the statistical patterns in the training data and produces new records that look realistic but belong to no real person [4], [5]. GANs and VAEs have worked well in medical imaging, physiological time series, and tabular clinical records [6], [7], [8]. The problem is that most published studies stop at checking how similar the synthetic data looks to the real data, using metrics like Fréchet Inception Distance or Jensen-Shannon divergence. Those metrics measure similarity. They say nothing about whether adding synthetic patients actually helps a classifier predict outcomes on real patients. That is the question that matters in practice, and it has drawn far less attention, especially for small tabular clinical datasets in IoMT settings [9], [10].

We tackle that question head on. The PBC complete-case dataset of 276 patients is split into 193 for training and 83 for held-out testing (stratified 70/30, seed 42). Three generative models are built from scratch in TensorFlow 2.15: a Vanilla GAN, a Conditional Tabular GAN that conditions generation on patient outcome (deceased vs. censored/transplanted), and a Tabular Variational Autoencoder. Two quality filters are applied before augmentation. The first removes synthetic records whose biomarker values fall outside Tukey fence bounds computed from real training patients. The second is a consensus voting step that accepts only synthetic patients all three models independently place in the same region of feature space, on the basis that cross-architecture agreement is a stronger signal of data quality than any single model.

Three contributions are made. First, three structurally different generative architectures are compared on the same liver

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cirrhosis dataset, all trained from scratch, giving a fair head-to-head comparison. Second, an IQR-based plausibility filter and a consensus voting mechanism are introduced as practical pre-augmentation quality controls. Third, the evaluation goes beyond distributional metrics to directly test classifier performance on a real held-out test set, using 5-fold cross-validation, bootstrap confidence intervals, and McNemar's exact test together.

Section II reviews related work. Section III describes the dataset, models, and experimental design. Section IV presents the results, and Section V concludes.

II. Related Work

A. Synthetic Data Generation for Healthcare

Synthetic data generation has drawn steady interest as a way to share healthcare datasets more widely without exposing patient privacy [4], [5], [11]. Gonçalves et al. reviewed generation and evaluation methods and argued that downstream utility, meaning how well synthetic data can stand in for real data in analysis, deserves as much attention as privacy [4]. Gonzales et al. surveyed clinical applications across oncology, cardiology, and infectious disease, and singled out tabular electronic health records as the area where good generation methods are most needed [5].

Classical augmentation approaches have well-documented limitations in this context. SMOTE addresses class imbalance by interpolating between existing minority-class training points and cannot generate samples outside the convex hull of observed data [12], [13]. Rule-based and parametric generators offer transparency but fail to capture the non-linear dependencies characterising clinical biomarker profiles. Deep generative models learn these dependencies directly from data and have become the dominant approach for tabular clinical synthesis [6], [7].

B. GAN Architectures for Tabular Clinical Data

Generative Adversarial Networks train a generator against a discriminator through adversarial feedback [14]. In clinical settings, Che et al. showed that GAN-augmented training data can improve risk prediction from electronic health records [8], and Choi et al. proposed medGAN for generating multi-label discrete patient records [15]. Standard GAN architectures are prone to mode collapse and training instability when applied to heterogeneous tabular data combining continuous biomarkers, binary indicators, and ordinal staging variables [16], [10]. The Conditional Tabular GAN addresses these limitations by conditioning both generator and discriminator on a class label, ensuring minority classes receive proportional representation [14], [17]. The Tabular Variational Autoencoder takes a complementary approach, optimising a probabilistic lower bound on the data likelihood. Its encoder maps patient records into a continuous latent space while its decoder reconstructs records from samples drawn from that distribution, generating new patients at inference time by sampling from a standard normal prior [7].

C. Synthetic Data for IoMT

IoMT infrastructures produce continuous streams of physiological measurements that clinical decision support systems must process reliably, requiring large labelled training sets [1], [2]. Meidan et al. showed that GAN-generated data improved anomaly detection in IoMT systems [18]. D'Amico et al. showed that AI-generated synthetic cohorts can speed up research in haematological conditions where real patient numbers are just as limited [19]. Kim et al. found that synthetic augmentation improves survival prediction in early-onset colorectal cancer, a rare-disease tabular setting close to PBC [20]. Tucker et al. showed that high-fidelity synthetic patient data can stand in for real data when evaluating healthcare software, as long as its quality is checked carefully [21].

D. From Distributional Fidelity to Predictive Utility

The dominant evaluation framework measures how closely generated samples resemble real data through marginal distributions and aggregate distance metrics [6], [13], [22]. Espinosa and Figueira found only a weak correlation between distributional and utility metrics across a range of datasets [10]. Foraker et al. showed that analytical results from synthetic data do not reliably replicate real-data findings even when distributional similarity is high [23]. Kababji et al. concluded that the ability to reproduce hazard ratios from real patient data is a more clinically meaningful benchmark than FID alone [9]. Kaabachi et al. identified the absence of held-out real-patient evaluation as a critical and recurring gap in the medical synthetic data literature [24]. The present work is designed specifically to fill that gap.

E. Multi-Model Ensemble Strategies and the Positioning of This Work

Jiang et al. proposed a multi-source data augmentation framework for industrial fault diagnosis in which several GAN models are trained independently, and their outputs are pooled through cooperative and competitive mechanisms [25]. The consensus voting mechanism introduced here shares this intuition but departs from it in two important respects: three structurally distinct architectures are used rather than multiple instances of the same one, and acceptance requires cross-architecture agreement rather than simply pooling all outputs.

Within the IoMT domain, Pandey et al. introduced MF-CGAN, a multifeature conditional GAN published in this journal that generates synthetic network traffic and health metrics by conditioning on multiple concurrent variables [26]. The present work complements MF-CGAN in the clinical direction: the data consists of continuous laboratory biomarkers, ordinal disease staging, and binary clinical indicators measured at a single visit. The evaluation criterion differs as well: classifiers are trained on augmented data and tested exclusively on real held-out patients. The consensus voting mechanism introduced here has no analogue in MF-CGAN, and our leakage-free evaluation shows that on this small clinical cohort it improves neither peak accuracy, cross-validation stability, nor disclosure risk—a negative result that is itself informative for the design of synthetic-augmentation pipelines.

III. METHODOLOGY

A. Dataset and Preprocessing

The dataset used in this study is the Mayo Clinic Primary Biliary Cirrhosis (PBC) trial [27], obtained from the UCI Machine Learning Repository [28]. The raw file contains 418 patient records and 20 columns. The patient identifier column carries no clinical information and is discarded. The remaining 19 columns consist of 11 continuous laboratory measurements (N_Days, Age, Bilirubin, Cholesterol, Albumin, Copper, Alk_Phos, SGOT, Triglycerides, Platelets, and Prothrombin), five binary clinical indicators (Sex, Drug, Ascites, Hepatomegaly, and Spiders), two ordinal variables (Edema coded as 0 = none, 1 = slight/suspected, 2 = confirmed; Stage coded 1–4), and the binary survival outcome Status (0 = censored/transplanted, 1 = deceased).

Patients 313–418 belong to an observational cohort missing between 9 and 10 laboratory measurements each. Rather than imputing values collected under fundamentally different protocols (which would introduce systematic bias), complete-case analysis is applied: any row containing at least one missing value is removed, yielding 276 complete patient records. Categorical columns are numerically encoded: Sex (Female = 0, Male = 1), Drug (Placebo = 0, D-penicillamine = 1), Ascites/Hepatomegaly/Spiders (No = 0, Yes = 1), and Status (censored/transplanted = 0, deceased = 1). All features are then scaled to $[0, 1]$ using a MinMaxScaler fitted exclusively on the training partition to prevent data leakage. The Pearson correlation structure across all 276 complete cases is given in Fig. S1; Bilirubin and N_Days emerge as the most clinically informative axes, with Bilirubin correlating positively with Copper ($r = 0.46$) and SGOT ($r = 0.42$), and N_Days correlating negatively with Bilirubin ($r = -0.43$).

The 276 records are divided into a training set of 193 patients and a held-out test set of 83 patients using stratified random splitting with seed 42, preserving class balance in both partitions (training: 115 survived/transplanted = 59.6%, 78 deceased = 40.4%). The overall pipeline is illustrated in Fig. 1.

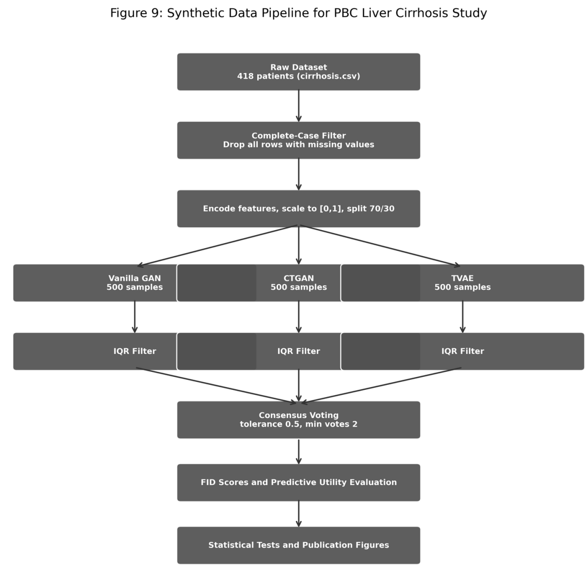

Fig. 1. End-to-end synthetic data pipeline. The raw 418-patient dataset passes through a complete-case filter and a preprocessing stage before the 193-patient training set feeds three parallel generative models. Each model's 500 generated records pass through an IQR plausibility filter before all three filtered pools enter the consensus voting stage.

B. Generative Model Architectures

All three models are implemented from scratch in TensorFlow 2.15 without third-party synthetic data libraries. Each model receives the MinMax-scaled training data and produces samples in the same normalised $[0, 1]$ range, which are inverse-transformed to recover original clinical units after generation. Each model generates 500 synthetic records.

Vanilla GAN. The generator maps a 100-dimensional noise vector $\mathbf{z}$ drawn from a standard normal distribution through two dense hidden layers of 128 and 256 units with Leaky ReLU activations ($\alpha = 0.2$) and batch normalisation (momentum $= 0.8$) to a sigmoid output of dimension 18. The discriminator maps each patient record through two dense layers of 256 and 128 units with Leaky ReLU activations and dropout ($p = 0.3$). Both networks use the Adam optimiser with learning rate 2×10^{-4} and $\beta_1 = 0.5$ for 500 epochs at batch size 32. The adversarial objectives are:

$$\mathcal{L}_D = -\mathbb{E}[\log D(\mathbf{x})] - \mathbb{E}[\log(1 - D(G(\mathbf{z})))] \tag{1}$$

$$\mathcal{L}_G = -\mathbb{E}[\log D(G(\mathbf{z}))] \tag{2}$$

Conditional Tabular GAN (CTGAN). This model is a conditional tabular GAN inspired by the conditioning strategy of Xu et al.; it does not implement the full CTGAN architecture (mode-specific normalisation and training-by-sampling) but adopts its central idea of conditioning generation on the outcome label. It extends the Vanilla GAN by conditioning both generator and discriminator on the patient outcome label $\mathbf{y}$, supplied as a two-dimensional one-hot vector. The generator receives the concatenation $[\mathbf{z} \| \mathbf{y}]$, and the discriminator receives $[\mathbf{x} \| \mathbf{y}]$. Training uses balanced mini-batches, drawing equal numbers of survivors and deceased patients at each step. Architecture and optimiser settings match the Vanilla GAN, with training for 300 epochs.

Tabular Variational Autoencoder (TVAE). The encoder maps each patient record through two dense layers of 128 and 64 units with ReLU activations to a mean vector $\boldsymbol{\mu}$ and log-variance vector $\log(\boldsymbol{\sigma}^2)$ of dimension 32. A latent sample is drawn using the reparameterisation trick:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\varepsilon} \cdot \exp(0.5 \cdot \log(\boldsymbol{\sigma}^2)), \tag{3}$$

where $\boldsymbol{\varepsilon}$ is drawn from a standard normal. The decoder maps $\mathbf{z}$ through two dense layers of 64 and 128 units with ReLU activations to a sigmoid output of dimension 18. The training objective is the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{TVAE}} = \mathbb{E}\big[\|\mathbf{x} - \hat{\mathbf{x}}\|^2\big] + \beta \cdot D_{\text{KL}}\big(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\big) \tag{4}$$

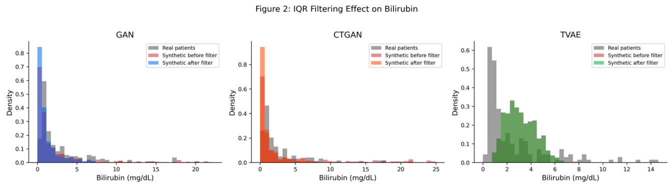

Fig. 2. Effect of IQR-based Tukey fence filtering on the Bilirubin distribution for all three generative models. Grey bars show real patient density. Pink bars show unfiltered synthetic distribution; coloured bars show the post-filter distribution. The GAN and CTGAN produce records extending beyond 15 mg/dL that the filter removes. The TVAE retains all generated records but produces a distribution shifted rightward relative to real patients.

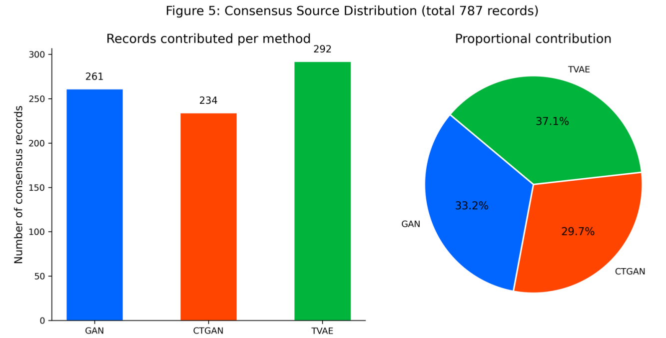

Fig. 3. Source composition of the 787 equalised consensus records. Left: absolute record counts per generative method after consensus voting with TVAE pool downsampled to 292. Right: proportional contribution as a pie chart. GAN contributes 33.2%, CTGAN 29.7%, and TVAE 37.1%.

with $\beta = 1.0$, Adam optimiser at learning rate 1×10^{-3}, batch size 32, and 300 training epochs. Training loss curves for all three models are given in Fig. S2. The adversarial losses settle without the oscillation that signals mode collapse, and the TVAE ELBO falls sharply before levelling off by roughly epoch 50, so all three models are trained to convergence.

C. IQR-Based Physiological Plausibility Filter

A Tukey fence filter is applied independently to each model's output. For each of the 11 continuous clinical features, the first and third quartiles, Q_1 and Q_3, are computed from the real training set alone. A synthetic record is retained only if every continuous feature value falls within $[Q_1 - 1.5 \times \text{IQR},\ Q_3 + 1.5 \times \text{IQR}]$. A single out-of-range value disqualifies the entire record. Binary and ordinal features are exempt because they are snapped to valid integer ranges during post-processing.

Applied to 500 generated records per model, the filter retained 331 GAN records (66.2%), 292 CTGAN records (58.4%), and 499 TVAE records (99.8%). Fig. 2 illustrates the effect on the Bilirubin distribution, the most challenging continuous feature to reproduce faithfully.

D. Multi-Method Consensus Voting

A consensus voting mechanism is applied that retains only records independently corroborated by all three generative models. This approach is related to the multi-source data augmentation strategy of Jiang et al. [25], who combine outputs from multiple independently trained GANs to reduce distribution skewness. The present work differs in that three structurally distinct architectures are used, and acceptance requires cross-architecture agreement rather than simply pooling all outputs.

A StandardScaler is fitted on the union of all three filtered datasets. For each candidate record from one model, the minimum Euclidean distance to any record from each of the other two models is computed in standardised space. The candidate is accepted if both other models have a record within tolerance $= 5.0$ standardised units. This process repeats with each filtered dataset as the candidate pool in turn, and exact duplicates are removed from the final set. The threshold of 5.0 was selected on the basis of a systematic sensitivity analysis conducted across eight values from 2.5 to 6.0. At thresholds of 3.0 and below, the accepted set contains around 10% deceased patients or fewer, a severe class imbalance relative to the real training cohort (40% deceased). As the tolerance increases the deceased fraction recovers (24.7% at 5.0) and the FID score falls monotonically (0.095 at 5.0), after which further increases yield only marginal FID improvements (0.0864 at 5.5, 0.0829 at 6.0) while admitting records with progressively weaker cross-model agreement. The value of 5.0 therefore represents the point at which class balance becomes acceptable and the quality improvement curve begins to flatten. To address the pool-size imbalance across the three filtered datasets (GAN 331 records, CTGAN 292 records, TVAE 499 records), the TVAE pool is randomly downsampled to 292 records before voting, matching the size of the smallest adversarial pool. This equalisation gives the two higher-fidelity adversarial models a proportionally stronger role in the consensus. The equalised consensus of 787 records achieves FID 0.0809, with source composition GAN 33.2%, CTGAN 29.7%, and TVAE 37.1%. Fig. 3 shows the source composition of the resulting set.

E. Synthetic Data Quality Evaluation

Distributional fidelity is assessed using a tabular proxy Fréchet Inception Distance computed on the 18-dimensional feature vectors:

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_s\|^2 + \text{Tr}\Big(\boldsymbol{\Sigma}_r + \boldsymbol{\Sigma}_s - 2(\boldsymbol{\Sigma}_r\boldsymbol{\Sigma}_s)^{1/2}\Big). \quad (5)$$

Lower values indicate greater distributional alignment. Per-feature Kolmogorov-Smirnov tests, Jensen-Shannon divergence, Cohen's d, Shapiro-Wilk normality tests, and chi-square tests for categorical features provide a granular distributional profile. Privacy risk is assessed through nearest-neighbour distance analysis. For each synthetic record, the Euclidean distance to the closest real training patient is computed in StandardScaler-normalised feature space across all 18 input features. A record is flagged as a near-duplicate if its distance falls below the 5th percentile of real-to-real nearest-neighbour distances, a data-driven threshold that identifies synthetic records closer to any real patient than 95% of real patients are to their nearest real neighbour.

TABLE I
SYNTHETIC DATA RETENTION AND FID SCORES

Method	Generated	Retained	Retention	FID
Vanilla GAN	500	331	66.2%	0.0995
CTGAN	500	292	58.4%	0.0724
TVAE	500	499	99.8%	0.2280
Consensus	—	787	—	0.0809

F. Predictive Utility Evaluation

Predictive utility is evaluated by training Random Forest (RF), Gradient Boosting (GB), and Logistic Regression (LR) classifiers under five training scenarios and testing on the 83-patient held-out real test set:

- **Scenario A:** 193 real patients only (baseline).
- **Scenario B:** 193 real + 292 IQR-filtered CTGAN records (485 total). CTGAN is selected here because it achieves the lowest FID among individual methods.
- **Scenario C:** 193 real + 787 consensus records (980 total).
- **Scenario D:** 193 real patients augmented with SMOTE oversampling (classical baseline, 230 total).
- **Scenario E:** 292 IQR-filtered CTGAN records only, with no real patients in training (synthetic-only transfer baseline).

The classifiers use fixed configurations (Random Forest and Gradient Boosting with 200 estimators, Gradient Boosting depth 3, Logistic Regression with 1000 iterations) applied identically across all scenarios so the comparison is fair. Evaluation metrics are Accuracy, F1, Precision, Recall, and AUC-ROC. Five-fold stratified cross-validation, 1000-resample bootstrap confidence intervals, and McNemar's exact test complement the single-split evaluation. In cross-validation the folds are drawn from the real training patients only; synthetic records are added exclusively to the training side of each fold and every validation fold contains real patients alone, so no synthetic record ever leaks into validation. All random seeds are fixed at 42, and the generative models are trained under fixed Python, NumPy, and TensorFlow seeds so the full pipeline is reproducible.

IV. RESULTS AND DISCUSSION

A. Synthetic Data Retention and IQR Filtering

Table I summarises IQR filter retention rates and FID scores. The GAN and CTGAN each reject roughly one third to one half of their generated records; the TVAE retains almost all. The consensus procedure accepts 787 records from the equalised filtered pools (TVAE downsampled to 292).

B. Distributional Fidelity

Fig. 4 shows the tabular FID scores for all four synthetic datasets. CTGAN achieves the lowest FID of 0.0724, followed by the Vanilla GAN at 0.0995. The TVAE produces an FID of 0.2280 despite retaining all generated records, confirming that smooth variational generation reduces distributional sharpness. The equalised consensus set scores 0.0809.

Three further views support this ranking and are provided as supplementary material. Fig. S3 compares feature distributions for six representative clinical variables: Albumin

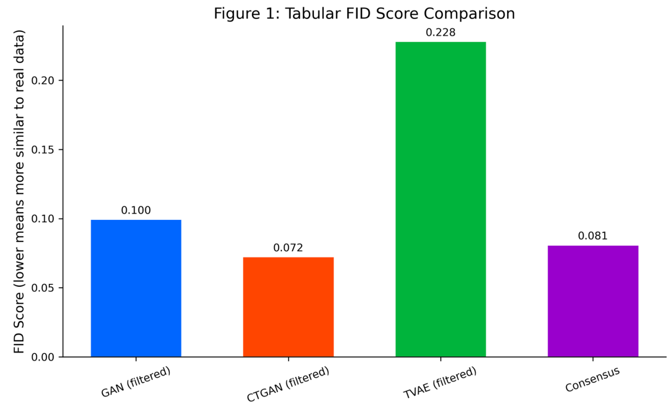

Fig. 4. Tabular Fréchet Inception Distance scores for all four synthetic datasets. Lower scores indicate greater distributional alignment with real training data. CTGAN achieves the lowest FID at 0.0724. The TVAE, despite retaining almost all generated records through the IQR filter, shows the highest FID at 0.2280, reflecting over-smoothing of clinical feature distributions. The equalised consensus set scores 0.0809, below the individual TVAE FID, confirming that pool equalisation improves distributional fidelity by reducing over-representation of variational samples.

and Prothrombin are well reproduced by GAN and CTGAN, while Bilirubin is the hardest feature, with the TVAE distribution shifted noticeably rightward relative to real patients. Fig. S4 compares the Pearson correlation structure of real and consensus data, showing that the consensus set preserves the dominant pairwise associations, including the positive Bilirubin-Copper and negative N_Days-Bilirubin relationships, though weaker correlations are attenuated. Fig. S5 gives a t-SNE projection in which the GAN and CTGAN sets mix with the real patient manifold, the TVAE points spread beyond the real density contours, and the consensus set concentrates in the central high-density region.

Per-feature KS tests show a consistent pattern. The filtered GAN shows no significant distributional difference from real training data on Age (KS $= 0.083$, $p = 0.348$), Triglycerides (KS $= 0.081$, $p = 0.380$), and Prothrombin (KS $= 0.092$, $p = 0.234$), but diverges significantly on Bilirubin (KS $= 0.413$), Alk_Phos (KS $= 0.506$), and Cholesterol (KS $= 0.308$). The TVAE diverges significantly on all 11 continuous features. The consensus set achieves substantially smaller KS statistics on the most challenging biomarkers: Bilirubin KS $= 0.269$ vs. GAN's 0.413; Alk_Phos KS $= 0.249$ vs. GAN's 0.506.

C. Privacy Risk Assessment

Fig. 5 shows the privacy risk analysis using nearest-neighbour distances in standardised feature space. The GAN and CTGAN distributions are centred well above the risk threshold (5th percentile of real-to-real distances), indicating that most adversarially generated records fall at a safe distance from any real patient. The TVAE and consensus distributions have substantial mass below the threshold, reflecting the broader coverage of the variational latent space and the fact that consensus acceptance pools records lying close to real patients. Table II gives the near-duplicate rates and mean nearest-neighbour distances for each method. Consensus voting does

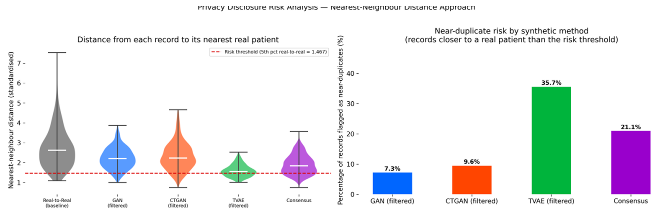

Fig. 5. Privacy disclosure risk analysis using nearest-neighbour distances in standardised feature space. Left: violin plots of distances from each synthetic record to its closest real training patient. The red dashed line marks the risk threshold at the 5th percentile of real-to-real distances. GAN and CTGAN distributions are centred above this threshold. Right: near-duplicate rates by method. GAN (7.3%) and CTGAN (9.6%) present low privacy risk, TVAE the highest at 35.7%, and the equalised consensus set 21.1%.

TABLE II
PRIVACY RISK ANALYSIS

Method	Records	Mean Dist.	Near-Dup.	Rate
GAN (filtered)	331	2.23	24	7.3%
CTGAN (filtered)	292	2.34	28	9.6%
TVAE (filtered)	499	1.60	178	35.7%
Consensus	787	1.91	166	21.1%

not improve on the individual adversarial pools. At a 21.1% near-duplicate rate it is far riskier than the filtered GAN (7.3%) or CTGAN (9.6%), because it inherits the TVAE records that sit closest to real patients.

D. Predictive Utility on the Held-Out Test Set

Table III reports complete test-set metrics across all five scenarios.

Fig. 6 shows the ROC curves for all three classifiers across all four scenarios on the 83-patient real held-out test set. All curves lie substantially above the random classifier diagonal. Gradient Boosting under Scenario A achieves the highest AUC at 0.878; the best augmented result is Logistic Regression under Scenario C at 0.850, which still does not exceed the baseline peak.

Because all of these numbers appear in Table III, the corresponding graphical views are supplied as supplementary material rather than repeated here. Fig. S6 gives Precision-Recall curves, which matter for this cohort given its class imbalance, and shows precision holding above 0.7 across a wide recall range for every classifier. Fig. S7 plots AUC across scenarios, Fig. S8 gives the F1 heatmap, and Figs. S9 and S10 compare all five metrics across scenarios. The pattern across all of them is the same: every scenario and classifier combination clears AUC $= 0.8$, so no augmentation strategy damages discrimination, but none uniformly improves on the baseline either. Gradient Boosting is the only classifier whose F1 falls monotonically as synthetic volume increases from Scenario A to C.

E. Cross-Validation

Table IV presents 5-fold stratified cross-validation AUC results, computed with synthetic records confined to the training side of each fold and every validation fold containing real patients only. Under this leakage-free protocol no augmentation scenario improves the mean cross-validated AUC over the real-data baseline: the baseline already reaches the highest Random Forest mean (0.884), and consensus augmentation (Scenario C) leaves the mean essentially unchanged for all three classifiers. The fold-to-fold standard deviation is likewise not meaningfully reduced: Gradient Boosting changes only from 0.071 in Scenario A to 0.054 in Scenario C (a 1.3-fold change) and Random Forest from 0.054 to 0.046 (a 1.2-fold change). This contrasts sharply with the large apparent variance reductions that arise if synthetic records are allowed into the validation folds, and shows that the earlier impression of a stability gain was an artefact of evaluating partly on synthetic data.

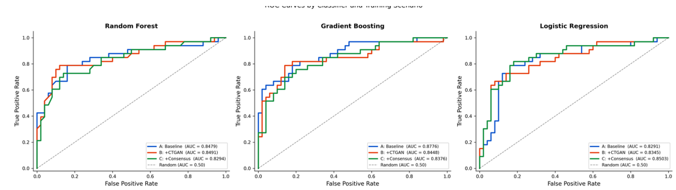

Fig. 6. ROC curves by classifier and training scenario on the 83-patient real held-out test set: Random Forest (left), Gradient Boosting (centre), Logistic Regression (right). All curves lie substantially above the random classifier diagonal. Gradient Boosting under Scenario A achieves the highest AUC at 0.878. Scenario D (SMOTE) achieves the highest overall accuracy at 0.807 under Gradient Boosting.

F. Bootstrap Confidence Intervals and McNemar's Test

Tables V and VI report 95% bootstrap confidence intervals for test-set AUC and McNemar's exact test results for the two deep-generative augmentation scenarios against the baseline. All confidence intervals overlap substantially across scenarios, and all McNemar p-values exceed 0.22, confirming that no augmentation scenario produces a statistically significant change in predictive accuracy over the real-data baseline on the 83-patient held-out test set.

Fig. 7 shows feature importance rankings from Random Forest and Gradient Boosting trained under Scenario A. Both classifiers consistently rank continuous laboratory biomarkers highest, with N_Days and Bilirubin leading across both methods, followed by Prothrombin and Albumin. Binary clinical indicators such as Ascites, Drug, Spiders, and Edema carry comparatively low weight. This pattern is clinically coherent and validates the integrity of the preprocessing and modelling pipeline.

TABLE III
TEST-SET CLASSIFICATION PERFORMANCE (83 REAL HELD-OUT PATIENTS)

Scenario	Classifier	n_{train}	Accuracy	F1	Precision	Recall	AUC
A: Baseline	Random Forest	193	0.7831	0.7848	0.7901	0.7831	0.8479
A: Baseline	Gradient Boosting	193	0.7831	0.7852	0.7955	0.7831	**0.8776**
A: Baseline	Logistic Regression	193	0.7711	0.7735	0.7940	0.7711	0.8291
B: Real + CTGAN	Random Forest	485	0.7831	0.7848	0.7901	0.7831	0.8491
B: Real + CTGAN	Gradient Boosting	485	0.7590	0.7616	0.7782	0.7590	0.8448
B: Real + CTGAN	Logistic Regression	485	0.7349	0.7377	0.7617	0.7349	0.8345
C: Real + Consensus	Random Forest	980	0.7349	0.7375	0.7477	0.7349	0.8294
C: Real + Consensus	Gradient Boosting	980	0.7229	0.7258	0.7457	0.7229	0.8376
C: Real + Consensus	Logistic Regression	980	0.7831	0.7854	0.8022	0.7831	0.8503
D: Real + SMOTE	Random Forest	230	0.7470	0.7496	0.7628	0.7470	0.8388
D: Real + SMOTE	Gradient Boosting	230	**0.8072**	**0.8087**	0.8140	0.8072	0.8661
D: Real + SMOTE	Logistic Regression	230	0.7711	0.7735	0.7940	0.7711	0.8297
E: Synthetic Only (CTGAN)	Random Forest	292	0.7108	0.7126	0.7556	0.7108	0.8230
E: Synthetic Only (CTGAN)	Gradient Boosting	292	0.6265	0.6281	0.6718	0.6265	0.6842
E: Synthetic Only (CTGAN)	Logistic Regression	292	0.6627	0.6634	0.7152	0.6627	0.8376

TABLE IV
5-FOLD CROSS-VALIDATION AUC (MEAN $\pm$ STD, REAL-ONLY VALIDATION FOLDS)

Scenario	RF	GB	LR
A: Baseline	$\mathbf{0.884 \pm 0.054}$	0.854 ± 0.071	0.852 ± 0.049
B: Real + CTGAN	0.865 ± 0.050	0.829 ± 0.063	0.847 ± 0.059
C: Real + Consensus	0.855 ± 0.046	0.833 ± 0.054	0.848 ± 0.056
D: Real + SMOTE	0.886 ± 0.045	0.860 ± 0.063	0.852 ± 0.052

TABLE V
95% BOOTSTRAP CONFIDENCE INTERVALS FOR TEST-SET AUC (1000 RESAMPLES)

Scenario	Classifier	AUC	95% CI
A: Baseline	RF	0.8479	[0.754, 0.938]
A: Baseline	GB	0.8776	[0.788, 0.948]
A: Baseline	LR	0.8291	[0.728, 0.916]
B: Real + CTGAN	RF	0.8491	[0.762, 0.930]
B: Real + CTGAN	GB	0.8448	[0.743, 0.930]
B: Real + CTGAN	LR	0.8345	[0.739, 0.917]
C: Real + Consensus	RF	0.8294	[0.727, 0.918]
C: Real + Consensus	GB	0.8376	[0.749, 0.919]
C: Real + Consensus	LR	0.8503	[0.754, 0.935]

TABLE VI
MCNEMAR'S EXACT TEST (AUGMENTED VS. BASELINE, $\alpha = 0.05$)

Classifier	Comparison	n_{10}	n_{01}	p-value
Random Forest	A vs. B	2	2	1.000
Random Forest	A vs. C	7	3	0.344
Gradient Boosting	A vs. B	4	2	0.688
Gradient Boosting	A vs. C	8	3	0.227
Logistic Regression	A vs. B	7	4	0.549
Logistic Regression	A vs. C	4	5	1.000

n_{10}: baseline correct, augmented wrong; n_{01}: augmented correct, baseline wrong.

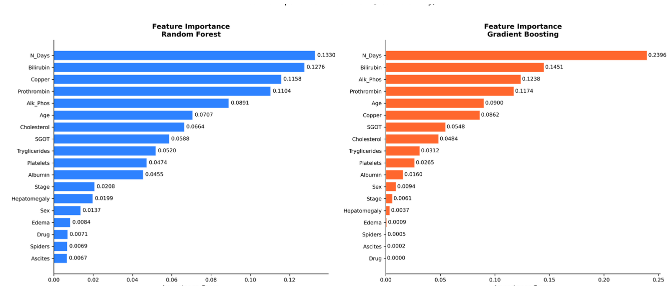

Fig. 7. Feature importance rankings from Random Forest (left) and Gradient Boosting (right) trained on real data only under Scenario A. Both classifiers consistently rank continuous laboratory biomarkers highest. N_Days and Bilirubin are the two leading predictors across both classifiers, followed by Prothrombin and Albumin. Binary clinical indicators show comparatively low importance, consistent with the dominant role of laboratory measurements in PBC prognosis.

G. Discussion

None of the augmentation scenarios produced a statistically significant improvement on the 83-patient held-out test set. This result is not surprising once you consider that Gradient Boosting already reaches AUC $= 0.878$ on real data alone. The PBC dataset carries enough signal for classifiers to perform well without synthetic help, and adding distributional approximations of the real patients does not move the decision boundary enough to change held-out predictions.

The cross-validation results reinforce this picture rather than overturning it. When training is repeated across five folds with synthetic data confined to the training side, no augmentation scenario improves the mean AUC over the baseline, and the fold-to-fold variance is essentially unchanged (Gradient Boosting 0.071 to 0.054, Random Forest 0.054 to 0.046). An earlier analysis that mixed synthetic records into the validation folds appeared to show a large variance reduction, but that was an evaluation artefact: the consensus records are selected from the dense centre of feature space and are therefore easier and more homogeneous to predict, so including them in validation mechanically shrank the measured variance. Once validation is restricted to real patients, the apparent stability benefit disappears. This is a cautionary result in its own right—synthetic augmentation must never be evaluated on folds that themselves contain synthetic data.

Fig. 8 quantifies the statistical power of the McNemar test for this dataset size. Assuming a discordant pair rate of around 15 per cent, the test has only 9% power to detect a genuine 5 percentage-point accuracy improvement. To reach 80% power at that effect size, roughly 300 held-out patients would be needed. This does not invalidate the null result—the p-values in this study ranged from 0.50 to 1.00, nowhere near the threshold—but it does mean the study cannot rule out a small real benefit. A replication on a larger cohort is needed before augmentation can be conclusively dismissed as unhelpful for this disease.

Scenario E in Table III evaluates whether a classifier trained exclusively on synthetic data can generalise to real

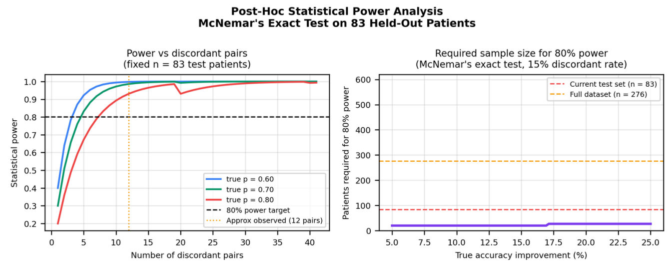

Fig. 8. Post-hoc power analysis for McNemar's exact test on 83 held-out patients. Left: statistical power versus number of discordant pairs for three assumed response proportions; with roughly 12 observed discordant pairs the test has below 9% power to detect a 5 percentage-point accuracy gain at $\alpha = 0.05$. Right: required test-set size for 80% power versus true accuracy improvement assuming a 15% discordant pair rate, showing that approximately 300 patients are needed.

patients. Using the 292 IQR-filtered CTGAN records as the sole training set and the held-out 83 real patients as the test set, the linear and bagging classifiers retain meaningful discriminative ability—Logistic Regression AUC $= 0.838$ and Random Forest AUC $= 0.823$—while Gradient Boosting is markedly more sensitive to the absence of real data, falling to AUC $= 0.684$. That two of the three classifiers exceed AUC 0.82 despite zero real-patient exposure during training shows the synthetic data carries transferable predictive signal, which is relevant for federated IoMT settings where real patient data cannot be shared across institutions but synthetic surrogates are available. The Gradient Boosting result is an important caveat: this transfer is not uniform across model families, and a boosting model in particular should not be trained on synthetic data alone.

Fig. 9 shows the subgroup consistency results. Classifiers trained under Scenario A and Scenario C were evaluated separately on three clinically meaningful splits of the test set: disease stage (early Stage 1–2, n = 17; late Stage 3–4, n = 66), sex (female n = 72; male n = 11), and treatment arm (D-penicillamine n = 39; placebo n = 44). For the subgroups large enough to produce reliable estimates—late stage, female, and both treatment arms—AUC differences between Scenario A and Scenario C stayed within about ± 0.06 across all classifiers, with no consistent direction of change, and every AUC exceeded 0.73. The null result therefore holds across the main clinical strata in this cohort, not just overall. The early-stage ($n = 17$) and male ($n = 11$) subgroups were too small to draw conclusions from and show erratic swings—for example, Random Forest AUC moves from 0.83 to 0.62 on the 11 male patients—so their results are reported only for completeness. External validation on an independent cohort is still needed.

Fig. 10 summarises the TVAE privacy analysis, which is relevant precisely because consensus voting does not by itself deliver a low-risk set (21.1% near-duplicates, Table II). The filtered TVAE pool of 499 records has a 35.7% near-duplicate rate, meaning more than one in three generated records sits closer to a real training patient than the 5th-percentile real-to-real distance threshold. To address this without retraining the model, Gaussian noise was added to the eleven continuous

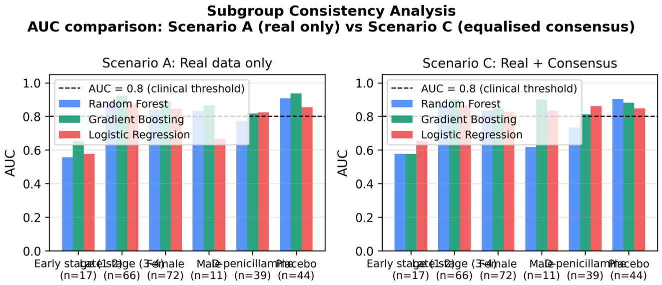

Fig. 9. Subgroup consistency analysis comparing Scenario A (real data only) and Scenario C (real + equalised consensus) across six clinical splits of the 83-patient test set. AUC differences between scenarios remain within about ± 0.06 for all subgroups large enough for reliable inference (late stage, female, both treatment arms), and every AUC exceeds 0.73. Early-stage ($n = 17$) and male ($n = 11$) subgroups are too small to draw conclusions from.

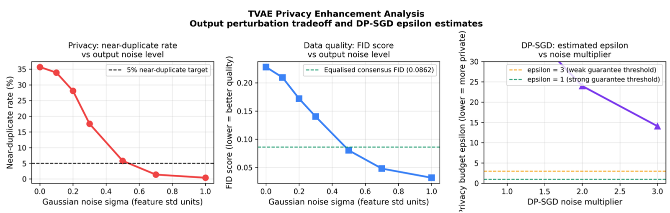

Fig. 10. TVAE privacy enhancement analysis. Left: near-duplicate rate versus Gaussian output noise σ (in feature standard-deviation units); at $\sigma = 0.7$ the rate falls from 35.7% to 1.4%, crossing the 5% near-duplicate target. Centre: tabular FID score versus σ; FID improves from 0.228 to 0.0481, below the equalised consensus benchmark of 0.0809. Right: estimated DP-SGD privacy budget ε versus noise multiplier at $\delta = 10^{-5}$; ε remains above 14 even at a multiplier of 3.0, confirming that formal differential privacy is not achievable at $n = 193$.

features of each TVAE record after generation, scaled relative to each feature's training-data standard deviation. At $\sigma = 0.7$, the near-duplicate rate drops from 35.7% to 1.4% and FID improves from 0.228 to 0.0481, which is better than the equalised consensus FID of 0.0809. The FID improvement is counterintuitive but makes sense: the TVAE was clustering records near specific real patients, so spreading those clusters out brings the marginal distributions closer to the real data overall. Formal DP-SGD training was also evaluated across noise multipliers from 0.5 to 3.0, producing epsilon values between 14.0 and 210.4 at $\delta = 10^{-5}$. These values are too large to provide any meaningful privacy guarantee. Output perturbation at $\sigma = 0.7$ is therefore the most practical mitigation available at $n = 193$. For deployments that require formal differential privacy, DP-SGD becomes viable only once a cohort of several hundred patients is available.

H. Why Augmentation Did Not Improve Prediction

The null result deserves a mechanistic explanation rather than being left as a bare empirical observation.

A generative model trained on the 193 real training patients is a function of those patients and a noise source. Whatever it emits is therefore a transformation of information already contained in the training set. By the data processing inequality, the synthetic records cannot carry more information about the true population distribution than the data they were derived from. Augmentation can reweight or smooth the existing training signal, but it cannot introduce fresh evidence about patients the model has never seen. That places a hard ceiling on what augmentation can deliver, and on a cohort where the baseline classifier already reaches AUC $= 0.878$, there is very little headroom beneath that ceiling to begin with.

The consensus mechanism works against itself in this respect. Cross-architecture agreement accepts, by construction, records that all three models place in the same region of feature space, and that region is the dense centre of the training distribution. The privacy results confirm it directly: the consensus set carries a 21.1% near-duplicate rate, higher than either adversarial pool, which means its records sit unusually close to real patients. Records close to existing training points are almost by definition redundant to a classifier. They reinforce parts of the feature space that are already well covered and add nothing near the decision boundary, which is the only place extra samples would change a fitted model.

This also accounts for the classifier-dependent pattern. Random Forest and Gradient Boosting partition feature space with axis-aligned splits, so points falling inside existing partitions leave split thresholds essentially untouched. Logistic Regression fits a single global boundary and is correspondingly the model most sensitive to the added probability mass, which is consistent with it showing the largest movement across scenarios in Table III.

Scenario E supplies the clearest evidence for this reading. When synthetic data is the only training data it is no longer competing against a stronger real signal, and it performs respectably at AUC 0.82 to 0.84 for Random Forest and Logistic Regression. Synthetic data can substitute for real data that is unavailable, but it cannot add to real data that is already sufficient. That distinction, substitution rather than augmentation, is the practical lesson of this study.

I. Threats to Validity

Several factors bound how far these findings generalise.

The study covers one cohort and one disease. PBC progresses slowly and has strong, well-characterised biomarkers, and the baseline classifiers exploit that structure efficiently. On a noisier condition, or one whose individual predictors are weaker, augmentation would have more room to help than it does here.

Complete-case analysis removed 142 of the 418 records. Patients 313 to 418 belong to an observational cohort for which 9 to 10 laboratory measurements were never collected, so their exclusion is driven by trial protocol rather than by patient characteristics. The analysed cohort is nevertheless the randomised trial population, and the results need not transfer to the observational arm.

Classifiers use default scikit-learn settings apart from the tree counts stated in Section III. Tuning might raise or lower the baseline, and a weaker baseline would leave more room for augmentation to show value. Tuning was deliberately withheld so that all five scenarios meet identical model capacity.

The FID reported here is a tabular proxy computed on 18-dimensional feature vectors, not the Inception-network FID used for images. It ranks methods meaningfully against one another on this dataset, but its absolute magnitude should not be compared with image-domain FID values in the literature.

The near-duplicate rate is a distance-based heuristic for disclosure risk, not a formal privacy guarantee. A record far from every training patient in Euclidean space may still be re-identifiable through auxiliary information, and the DP-SGD analysis shows a formal guarantee is out of reach at this sample size.

Finally, the conditional tabular GAN implemented here adopts the outcome-conditioning strategy of Xu et al. but not the full published architecture, so these results should not be read as a benchmark of CTGAN as originally specified.

J. Practical Guidance for IoMT Deployments

Four recommendations follow for practitioners building clinical decision support on small tabular cohorts.

First, never evaluate augmentation on validation folds that contain synthetic records. The variance reduction reported in our own earlier analysis disappeared completely once validation was restricted to real patients. Any stability or accuracy gain measured on mixed folds should be treated as an artefact until it is reproduced on real-only validation.

Second, judge synthetic data on held-out real patients rather than on distributional metrics. The conditional tabular GAN achieved the best FID in this study and still delivered no predictive benefit. FID, KS statistics, and Jensen-Shannon divergence describe how the data looks; only downstream evaluation on real patients establishes whether it works.

Third, where synthetic data is genuinely required, for cross-site sharing or where real records cannot leave an institution, the conditional tabular GAN is the most practical choice in this setting, pairing the lowest FID with a low near-duplicate rate. Variational models should not be released without mitigation: the TVAE reached a 35.7% near-duplicate rate, which output perturbation at $\sigma = 0.7$ reduces to 1.4% while also improving FID.

Fourth, do not assume synthetic-only training transfers across model families. Random Forest and Logistic Regression held AUC above 0.82 with no real training data at all, but Gradient Boosting fell to 0.684. Any federated deployment that trains on synthetic surrogates should validate the specific model family it intends to ship.

V. Conclusion

This paper set out to answer whether synthetic patient data can improve liver cirrhosis survival prediction in an IoMT setting. The short answer is no: augmentation helped neither on the held-out test set nor in leakage-free cross-validation, and consensus voting did not improve the privacy profile. What the synthetic data does provide is transferable predictive signal when used on its own, and a clear methodological lesson about how such data must be evaluated.

Three generative models were trained from scratch, under fixed random seeds, on the Mayo Clinic PBC dataset. An IQR filter discarded roughly 34 to 42 per cent of generated records for clinical implausibility. A consensus voting step then accepted only records corroborated by all three architectures, yielding an equalised set of 787 records.

The main findings are as follows. CTGAN produced the best distributional quality at FID $= 0.072$, and the filtered adversarial pools carried the lowest disclosure risk (7.3 per cent near-duplicates for GAN, 9.6 per cent for CTGAN), making CTGAN the most practical single method for IoMT augmentation under privacy constraints. No augmentation scenario beat the real-data baseline by a statistically significant margin on the 83-patient test set, and once cross-validation was performed without leaking synthetic records into the validation folds, augmentation produced no improvement in mean AUC and no meaningful reduction in variance. Consensus voting did not lower disclosure risk, reaching a 21.1 per cent near-duplicate rate because acceptance pools records lying close to real patients. Classifiers trained on synthetic CTGAN data alone—no real patients in the training set—still achieved AUC of 0.82 to 0.84 for Random Forest and Logistic Regression on the real held-out test set (though Gradient Boosting fell to 0.68), confirming that the synthetic data carries genuine, if model-dependent, predictive signal that could be useful in federated IoMT settings where patient data cannot be shared across sites.

Two limitations constrain these findings. The McNemar test is underpowered at $n = 83$; roughly 300 held-out patients are needed to detect a 5-point accuracy gain with 80% power, so a small real benefit cannot be ruled out. TVAE near-duplicate risk falls from 35.7% to 1.4% with output perturbation at $\sigma = 0.7$, but formal differential privacy via DP-SGD is not viable until the cohort exceeds several hundred patients. Replication on a larger multi-site hepatological dataset and extension to longitudinal IoMT data streams remain the priority next steps.

Supplementary Material

Ten supporting figures are provided as supplementary material and are cited in the text as Fig. S1 to Fig. S10. They are: S1, the Pearson correlation matrix across all 276 complete cases; S2, training loss curves for all three generative models; S3, feature distribution comparisons for six representative clinical variables; S4, real versus consensus correlation structure; S5, a t-SNE projection of real and synthetic records; S6, Precision-Recall curves; S7, AUC across scenarios; S8, the F1-score heatmap; and S9 and S10, grouped comparisons of all five evaluation metrics across scenarios. Every quantity plotted in S6 to S10 also appears numerically in Tables III and IV. The supplementary figures are distributed with the code repository.

Ethics Statement

This study used the Mayo Clinic PBC dataset, which is publicly available, fully de-identified, and distributed for research use through the UCI Machine Learning Repository [28]. No new patient data were collected and no additional ethical approval was required.

Data and Code Availability

The Mayo Clinic PBC dataset is publicly available from the UCI Machine Learning Repository [28]. All generative models, filtering and consensus code, and evaluation scripts are implemented in Python and TensorFlow 2.15 and run under fixed random seeds (Python, NumPy, and TensorFlow all seeded at 42), so the full pipeline, including synthetic data generation, reproduces the numbers reported here on re-execution. The code is available at https://github.com/Michaeludousoro/liver-cirrhosis-synthetic-data.

Declaration of Generative AI Use

The authors used a generative AI assistant to help with code review, debugging, and language editing. All experiments, results, and scientific conclusions were produced and verified by the authors, who take full responsibility for the content of this paper.

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